

Day0100 -- CCTV and P2P Network

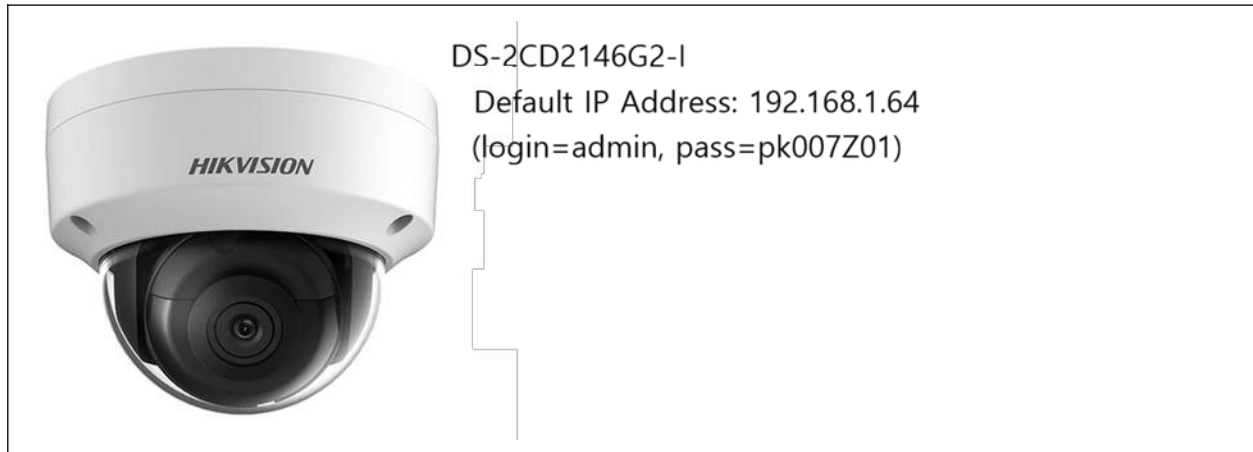
เจ้าของงาน ID:B6728960 Name:มงคล น้อย
บาท

คู่ทดลอง ID: B6729615 Name:ปภังกร สมสุข

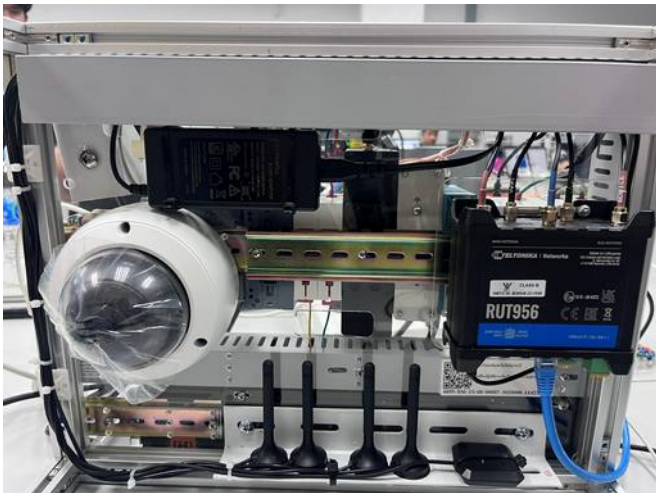
- Download >> https://drive.google.com/drive/folders/1ha-3NwO1UugEgqZGvxTNIUcfa5-sMf8u?usp=drive_link
- ทำการทดลอง เก็บข้อมูล ลองคนต่อกลุ่ม แล้วแยกกันเติมรายละเอียดรายงานให้เรียบร้อย
- กำหนดชื่อไฟล์ตามรูปแบบ "B3601234 Wichai Srisuruk – Day0100 CCTV and P2P Network.pdf"
- ส่งแบบ pdf รอบโครงร่าง ก่อน 20260808-1700 ที่ <https://forms.gle/wt2E2RnpLukV4qsW9>
- ส่งแบบ pdf รอบสมบูรณ์ ก่อน 20260830-0600 ที่ <https://forms.gle/wt2E2RnpLukV4qsW9>

Mission 1: IP Camera System

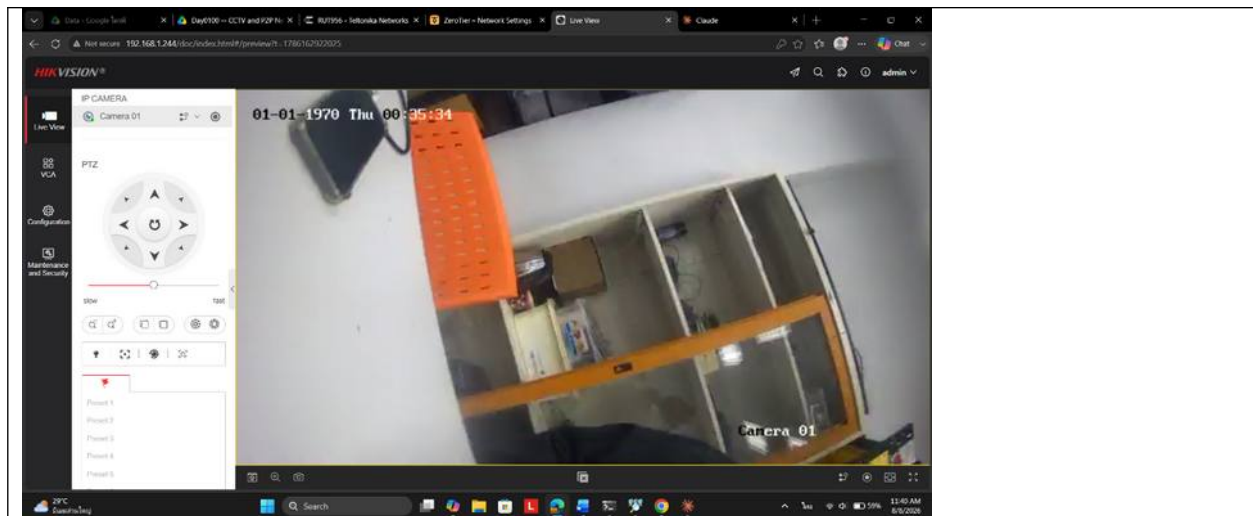
Mission 1/1: Capture จาก Web URL



รูปถ่ายวงจร - 1/1



แดปหน้าจการทำงาน - 1/2

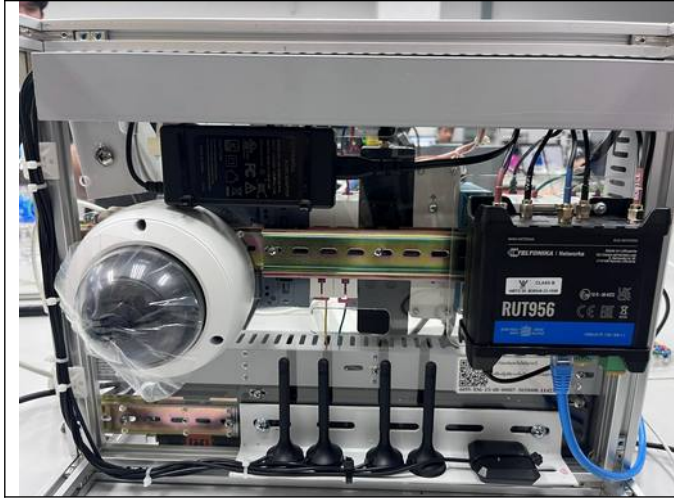


แดปหน้าจอการทำงาน - 2/2

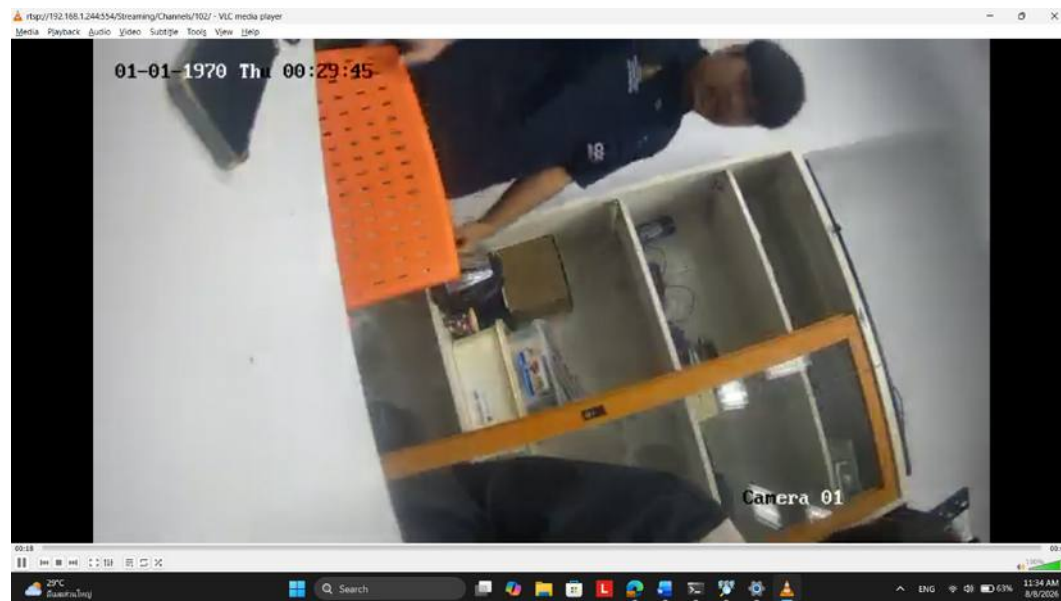


Mission 1/2: Capture จาก VLC Program

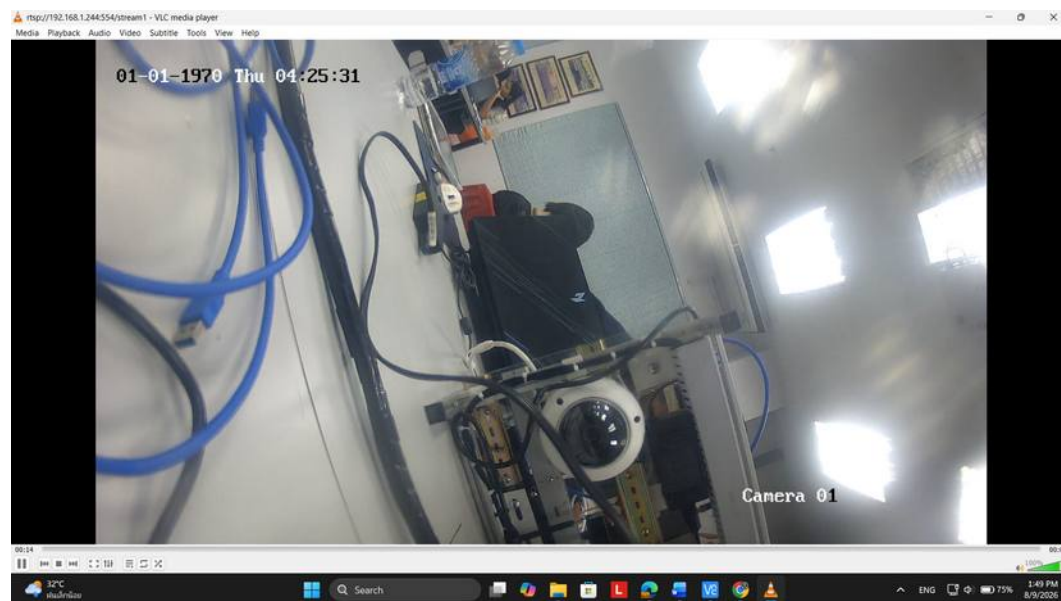
รูปถ่ายวงจร - 1/1



แดปหน้าจอการทำงาน – 1/2

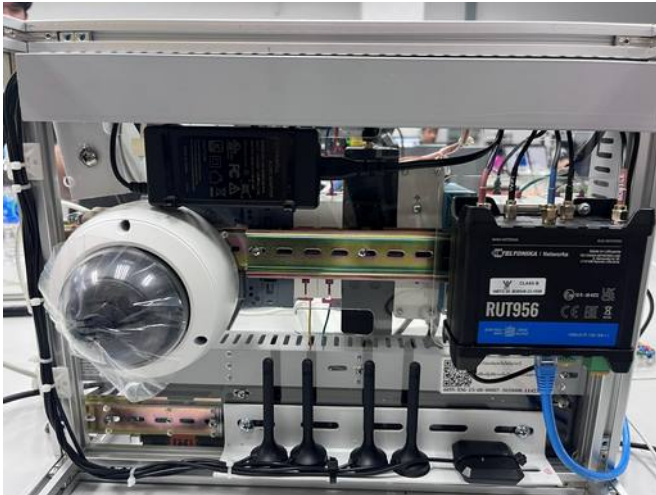


แดปหน้าจอการทำงาน – 2/2

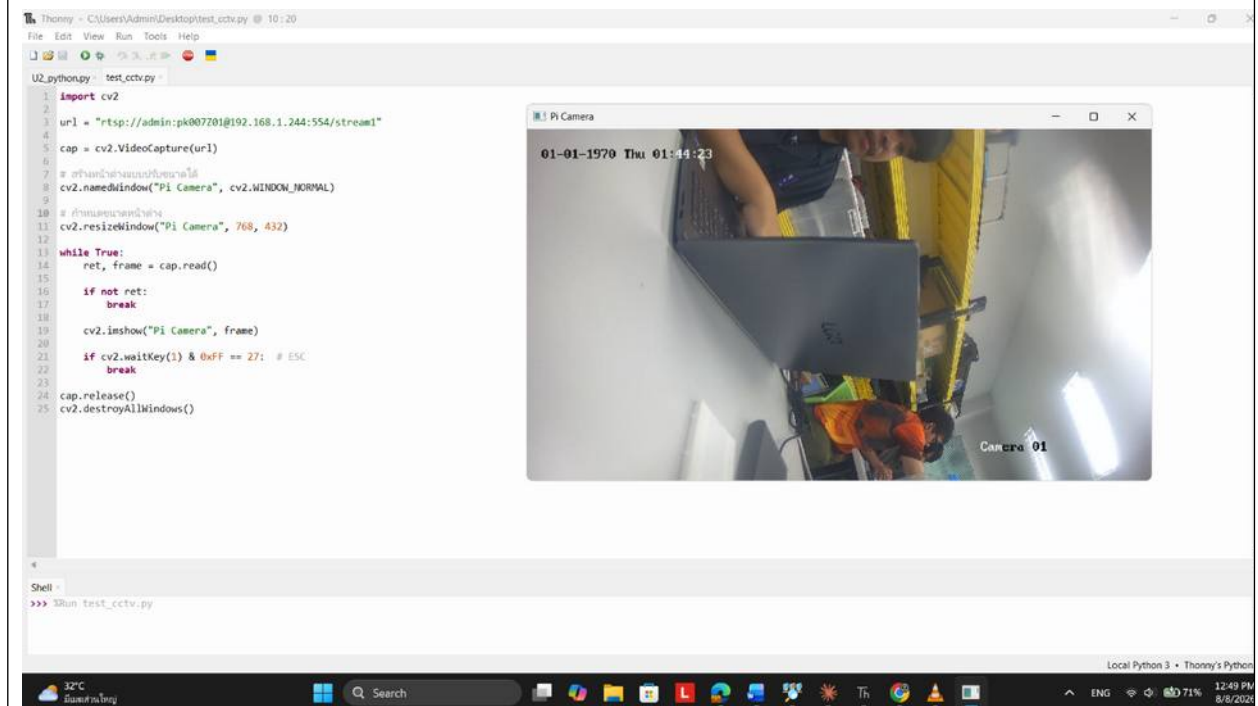


Mission 1/3: Capture จาก Python Code

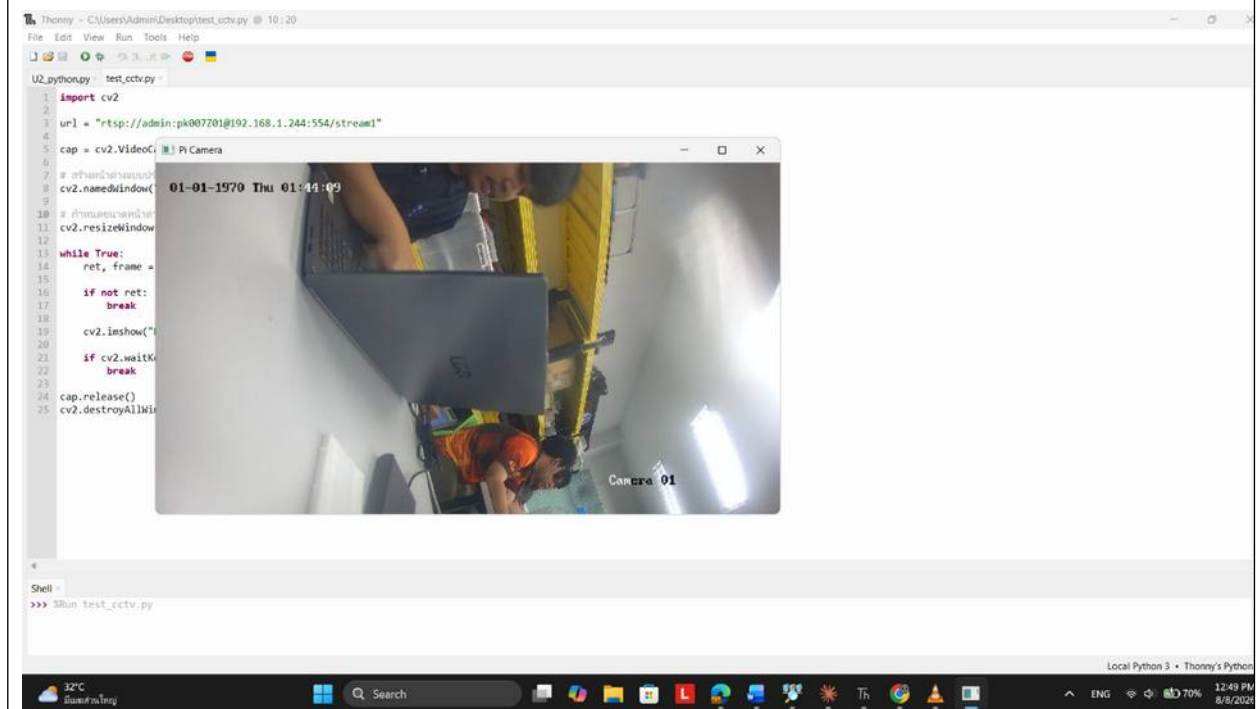
รูปถ่ายวงจร - 1/1



แดปหน้าจอการทำงาน - 1/2



แดปหน้าจอการทำงาน – 2/2



Mission2: RUT956 router with P2P Network

Mission 2/1: Remote On/Off Control in LAN Group



DHCP Server

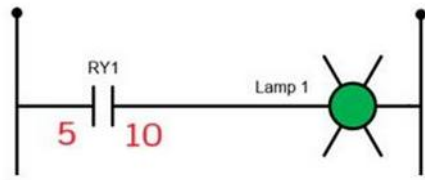
192.168.2.X

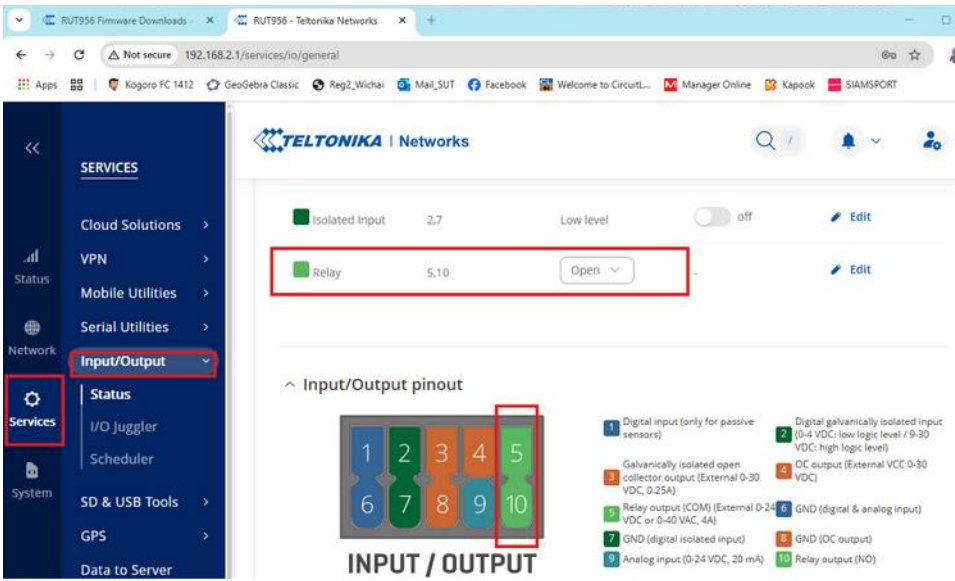
LAN1

RUT956

192.168.2.1

WAN





Services

Cloud Solutions

VPN

Mobile Utilities

Serial Utilities

Input/Output

Status

I/O Juggler

Scheduler

SD & USB Tools

GPS

Data to Server

TELTONIKA Networks

Isolated Input 2,7 Low level off Edit

Relay 5,10 Open Edit

Input/Output pinout

1 Digital input (only for passive sensors)

2 Digital galvanically isolated input (0-4 VDC: low logic level / 9-30 VDC: high logic level)

3 Galvanically isolated open collector output (External 0-30 VDC, 0.25A)

4 DC output (External VCC 0-30 VDC)

5 Relay output (COM) (External 0-24 VDC or 0-40 VAC, 4A)

6 GND (digital & analog input)

7 GND (digital isolated input)

8 GND (DC output)

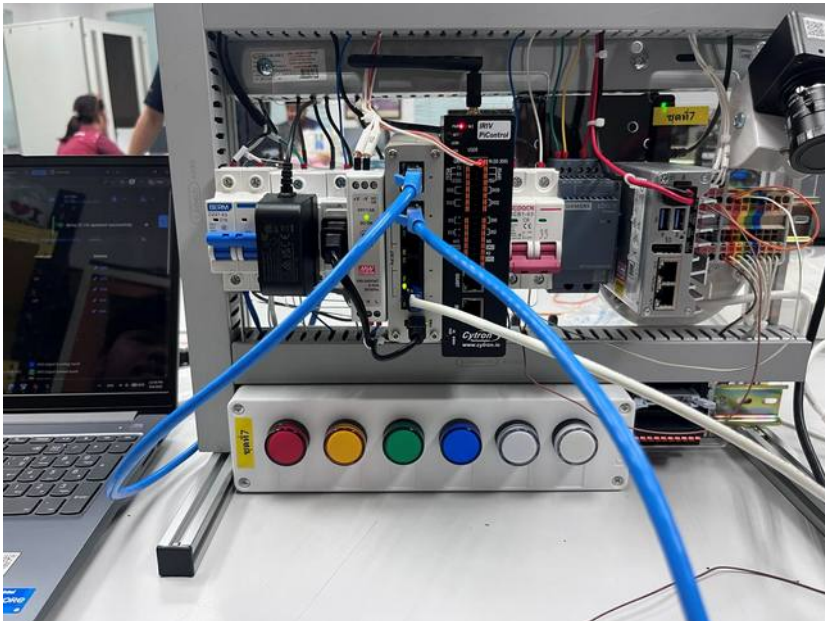
9 Analog input (0-24 VDC, 20 mA)

10 Relay output (NO)

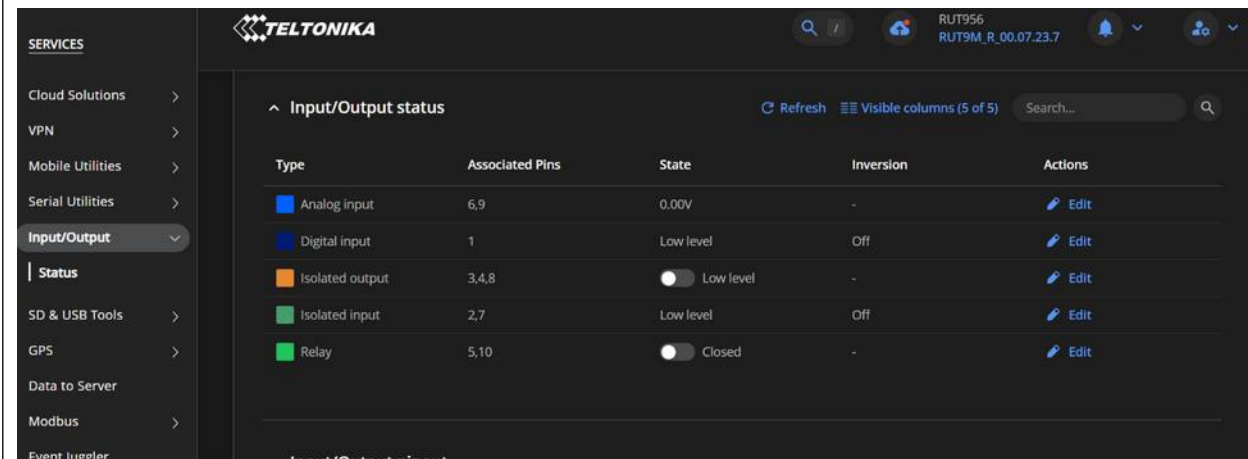
รูปถ่ายวงจร - 1/2



รูปถ่ายวงจร - 2/2



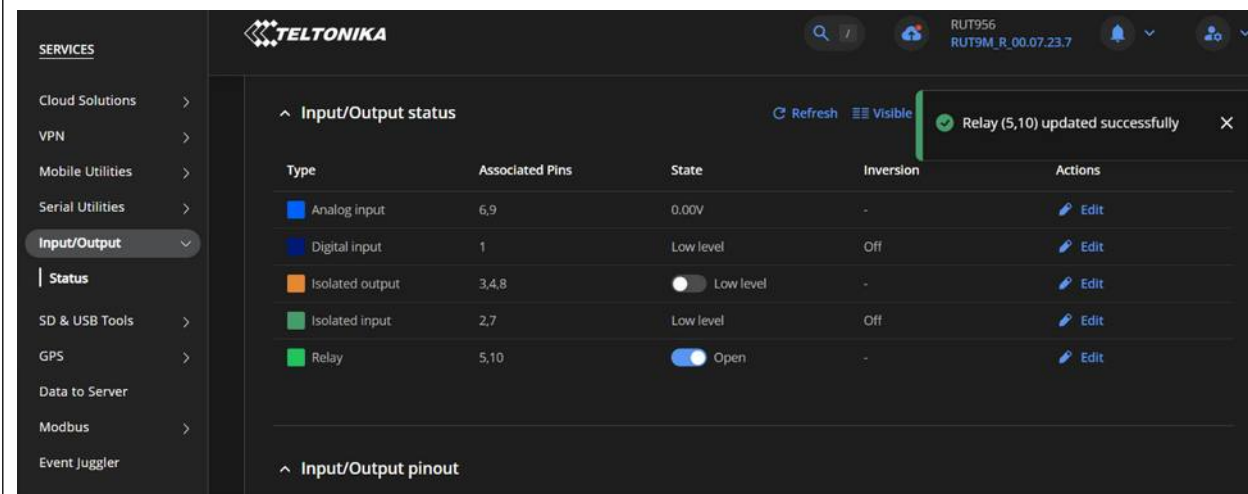
แดปหน้าจการทำงาน – 1/2



Input/Output status

Type	Associated Pins	State	Inversion	Actions
Analog input	6,9	0.00V	-	Edit
Digital input	1	Low level	Off	Edit
Isolated output	3,4,8	Low level	-	Edit
Isolated input	2,7	Low level	Off	Edit
Relay	5,10	Closed	-	Edit

แดปหน้าจการทำงาน – 2/2



Input/Output status

Relay (5,10) updated successfully

Type	Associated Pins	State	Inversion	Actions
Analog input	6,9	0.00V	-	Edit
Digital input	1	Low level	Off	Edit
Isolated output	3,4,8	Low level	-	Edit
Isolated input	2,7	Low level	Off	Edit
Relay	5,10	Open	-	Edit

Input/Output pinout

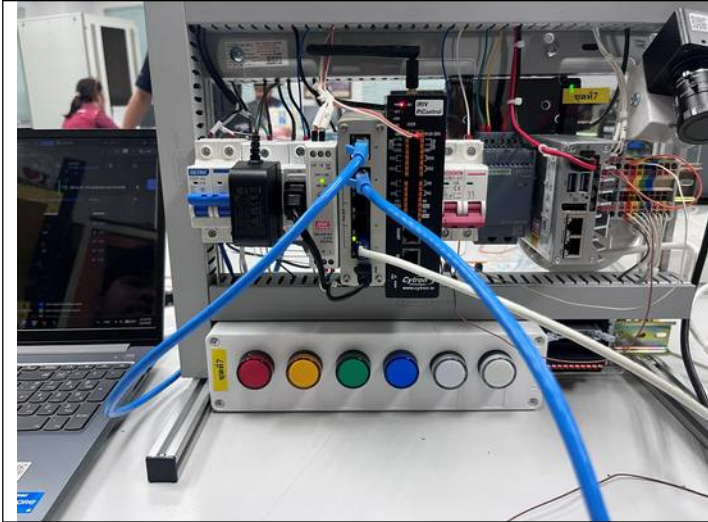
Mission 2/2: Remote On/Off Control in WAN Group

The diagram illustrates a ZeroTier network setup. A Teltonika RUT956 router, acting as a DHCP Server (192.168.2.1), is connected to a laptop (192.168.191.120) and a server (192.168.191.6). Both devices are part of a ZeroTier network, indicated by the ZeroTier logo and IP addresses. A globe icon represents the WAN connection.

The screenshot shows the ZeroTier Central web interface. The left sidebar lists various services, with 'Input/Output' highlighted. The main panel displays the 'Relay' status, which is 'Open'. Below this, the 'Input/Output pinout' is shown, detailing the functions of the 10 pins:

- 1: Digital input (only for passive sensors)
- 2: Digital galvanically isolated input (0-4 VDC low logic level / 9-30 VDC high logic level)
- 3: Galvanically isolated open collector output (External 0-30 VDC, 0.25A)
- 4: OC output (External VCC 0-30 VDC)
- 5: Relay output (COM) (External 0-24 VDC or 0-40 VAC, 4A)
- 6: GND (digital isolated input)
- 7: GND (digital isolated input)
- 8: GND (analog input)
- 9: GND (OC output)
- 10: Relay output (NO)

รูปถ่ายวงจร - 1/2



รูปถ่ายวงจร - 2/2



แดปหน้าจอการทำงาน - 1/2

Group 1

Day0100 - CCTV and P2P Network

RUT956 - Teltonika Networks

ZeroTier Central - Networks - 830

RUT956 - Teltonika Networks

Log in

Not securehttps://192.168.191.114/services/fo/general

Chat

<<

SERVICES

Cloud Solutions >

VPN >

Mobile Utilities >

Serial Utilities >

Input/Output >

Status

SD & USB Tools >

GPS >

Data to Server

Modbus >

Event Juggler

TELTONIKA

Search

RUT956

RUT9M_R_00.07.23.7

Notification

Users

Services > Input/Output > Status

Input/Output status

Refresh

Visible columns (5 of 5)

Search...

Type	Associated Pins	State	Inversion	Actions
Analog input	6,9	0.00V	-	Edit
Digital input	1	Low level	Off	Edit
Isolated output	3,4,8	Low level	-	Edit
Isolated input	2,7	Low level	Off	Edit
Relay	5,10	Open	-	Edit

Input/Output pinout

1

2

3

4

5

1 Digital input (only for passive sensors)

2 Digital galvanically isolated input (0-4 VDC: low logic level / 9-30 VDC: high logic level)

3

4 GND (digital & analog input)

5 GND (digital isolated input)

28°C

Basement 7th floor

Search

ENG

71%

10:18 AM

8/9/2024

แดปหน้าจการทำงาน - 2/2

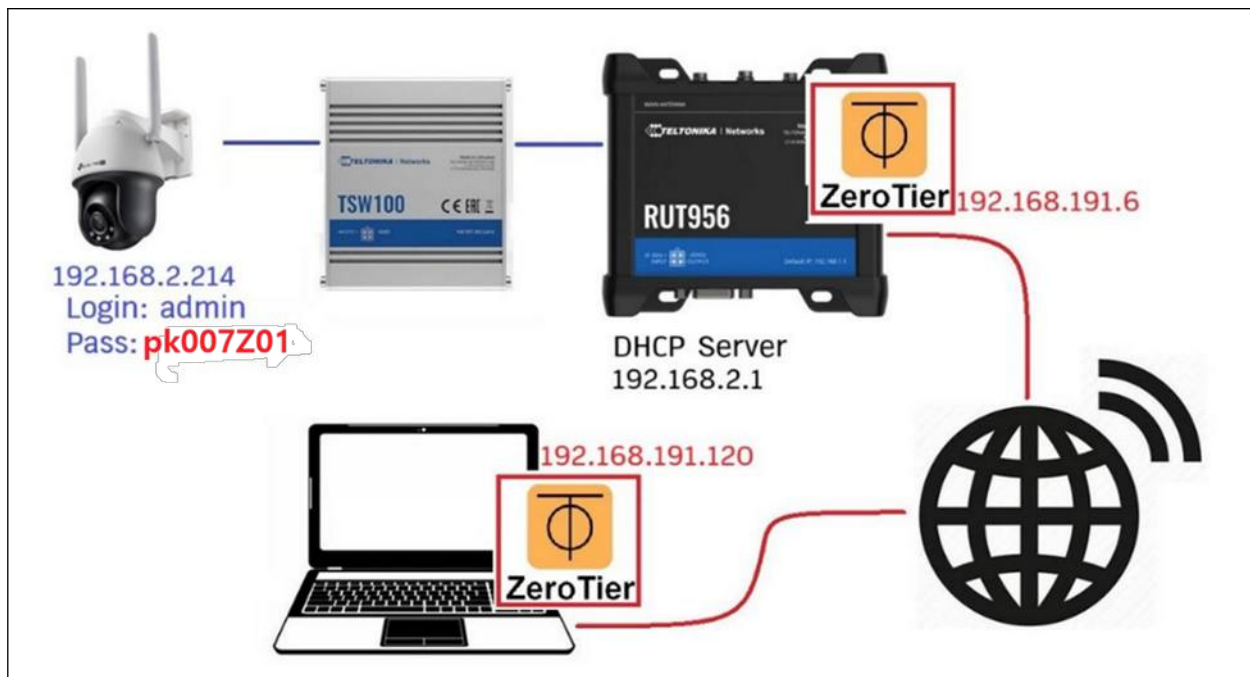
The screenshot displays the Teltonika RUT956 web interface. The left sidebar shows the 'Services' menu with 'Input/Output' selected. The main content area is titled 'Input/Output status' and features a table of device settings. A notification at the top right states 'Relay (5,10) updated successfully'. Below the table is the 'Input/Output pinout' section, which includes a diagram of the 10-pin header and a list of pin functions.

Type	Associated Pins	State	Inversion	Actions
Analog input	6,9	0.00V	-	Edit
Digital input	1	Low level	Off	Edit
Isolated output	3,4,8	Low level	-	Edit
Isolated input	2,7	Low level	Off	Edit
Relay	5,10	Closed	-	Edit

Input/Output pinout

- 1 Digital Input (only for passive sensors)
- 2 Digital galvanically isolated input (0-4 VDC; low logic level / 9-30 VDC; high logic level)
- 3 Galvanically isolated open collector output (External 0-30 VDC, 0.25A)
- 4 OC output (External VCC 0-30 VDC)
- 5 Relay output (N/O)
- 6 GND (digital & analog input)
- 7 GND (digital isolated input)
- 8 GND (OC output)
- 9 Analog input (0-24 VDC, 20 mA)
- 10 Relay output (N/O)

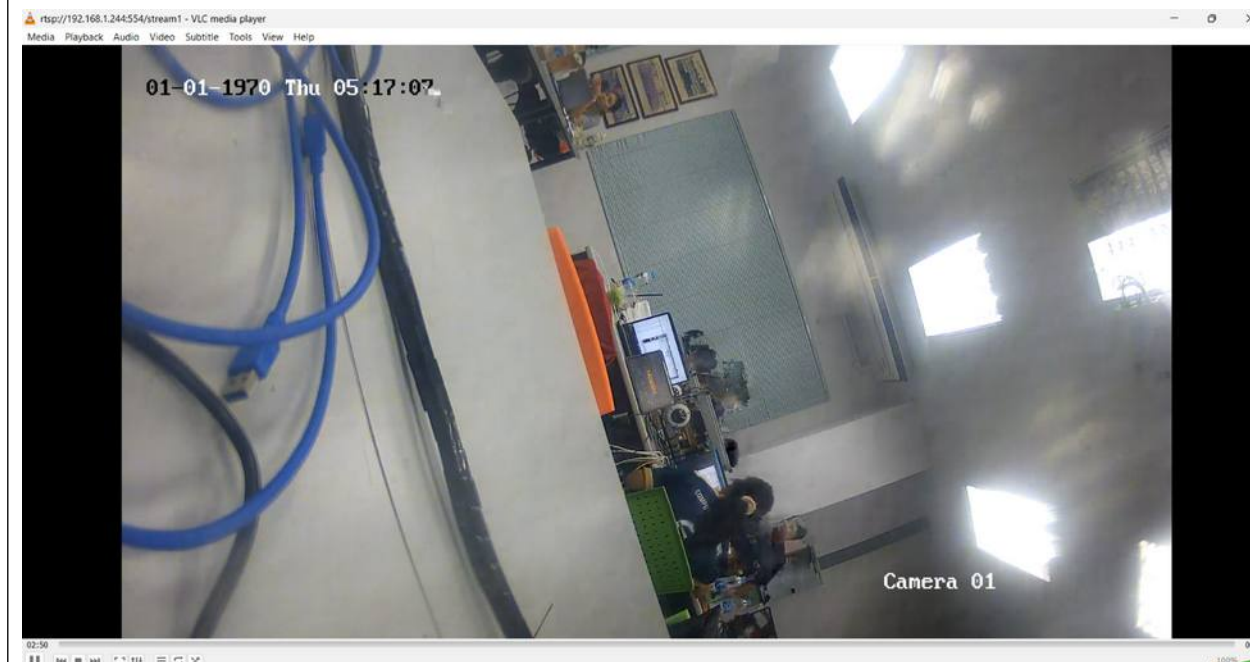
Mission 2/3: Remote Access IP Camera in WAN Group



รูปถ่ายวงจร - 1/2

รูปถ่ายวงจร - 2/2

แคปหน้าจอการทำงาน - 1/2



แดปหน้าจอการทำงาน - 2/2

